HCDE 493: Senior Capstone

ABET EAC Criterion 5b — Course Evaluation

Spring 2025 — 5 Quarter Credits — Prepared April 2026

Course	HCDE 493: Senior Capstone
Term	Spring 2026
Credits	5 quarter credits (QC) = 3.33 semester credits (SC)
Export file	hcde-493-a-sp-26-senior-capstone-export.imscc
Rubrics in export	Yes — 2 rubrics: Process Book Rubric (100 pts, 6 criteria) and Milestone Deliverable Rubric (15 pts, 3 criteria)

Criterion 5b Credit Estimates

Engineering Design	3.0–4.0 QC
Confidence	High
Engineering Science	0.0–0.5 QC
Confidence	Low
Total 5b (SC, low est.)	2.0 SC
Total 5b (SC, high est.)	3.0 SC

Note: 5 QC × 0.667 = 3.33 SC maximum. Credit estimates reflect the portion of course content qualifying under Criterion 5b, not the course's full credit value.

Course Description

HCDE 493 Senior Capstone is a 5-credit course described in the syllabus as 'the culminating capstone design experience in the HCDE major. It integrates knowledge and skills acquired during the program into a comprehensive project.' It is the execution phase of a two-quarter capstone sequence; students entering 493 have already produced a project proposal and methodology specification in HCDE 492 (Capstone Planning). In 493, teams execute the planned design process over 10 weeks with a client or sponsor, producing milestone reports, a poster, a video, a public showcase presentation, and a process book.

The course is structured around a small number of lectures with class time devoted primarily to group work and critique sessions with peers, instructors, and external reviewers. Three milestone deliverables (P1, P2, P3) report progress through each planned phase of the design project. Two weekly status reviews (S1, S2) and three rounds of peer feedback (PF1, PF2, PF3) constitute the administrative category. Final deliverables (F1–F5) document and present the completed project.

Course objectives include: developing a project vision from a sponsor's brief; choosing appropriate methodologies; considering the effects of design decisions on stakeholders; managing client relationships; communicating work professionally; and producing a process book that documents the full design process.

Module and Assignment Structure

The course has 12 Canvas modules across 10 weeks. Graded assignments fall into three categories:

Assignment	Description	Pts	ABET Relevance
Admin category (15% of grade)			
P0: Revised Project Plan	Update the 492 proposal plan to reflect new information at the start of the quarter	5	Engineering design — baseline for execution
S1: Weekly Status Review 1	Team progress report: milestone status, Gantt chart link, identification of 3+ project risks with mitigation plans	16	Engineering design — iterative project management and risk analysis
S2: Weekly Status Review 2	Same structure as S1; captures later-quarter status	16	Engineering design — continued risk tracking
PF1: Peer Feedback 1	Structured peer review of another team's milestone work	5	Engineering design — iterative critique loop
PF2: Peer Feedback 2	Structured peer review of another team's milestone work	5	Engineering design — iterative critique loop
PF3: Peer Feedback 3	Structured peer review of another team's milestone work	5	Engineering design — iterative critique loop
Project Process P* (25% of grade)			
P1: Milestone 1	Milestone report: what the team set out to do, how they did it, what they learned, and how it informs next steps	15	Primary engineering design — design execution, analysis, and synthesis
P2: Milestone 2	Same structure as P1; covers second phase of the design process	15	Primary engineering design — iterative execution
P3: Milestone 3	Same structure; covers third phase	15	Primary engineering design — iterative execution
Final Deliverables F* (60% of grade)			
F1: Project Poster	Visual summary of project: problem, process, and outcomes (20 pts)	20	Engineering design — design communication artifact
F2: Project Video	Short video documenting the project (20 pts)	20	Communication only — not engineering design
F3: Showcase Event	Public presentation at the annual HCDE Capstone Showcase (10 pts)	10	Communication only — not engineering design
F4: Process Book	Comprehensive documentation of the full design process: concept specification, design specification,	50	Primary engineering design — design specification and process documentation

	process comprehensiveness, process narrative, reflection (50 pts)		
F5: Final Reflection	Individual written reflection on learning and project experience (5 pts)	5	Reflective writing — not engineering design

Engineering Design

Credit Estimate: 3.0–4.0 QC | Confidence: High

HCDE 493 is the strongest engineering design course in the HCDE BS curriculum. The entire course structure is organized around executing a multi-phase, client-sponsored human-centered design project from initial planning through final deliverables. Students apply methods they specified in HCDE 492 to real design problems, produce iterative milestone reports, receive structured critique, revise their approach, and document the complete design process in a rubric-assessed process book.

The ABET EAC definition of engineering design is an iterative, creative, decision-making process that involves identifying opportunities, developing requirements, performing analysis and synthesis, generating and evaluating solutions, considering constraints, and making trade-offs. All of these elements are present in HCDE 493, supported by rubric criteria in the export.

Primary Evidence: P1, P2, P3 — Milestone Deliverables

The three milestone assignments (P1, P2, P3; 15 pts each) are the primary design execution evidence. Each milestone report requires students to explain what the team set out to do, how they did it, what they learned, and how findings inform next steps. The Milestone Deliverable Rubric assesses:

Criterion	Points / ABET Relevance
Completeness (6 pts): The milestone report addresses all required components: what work was done, the approach taken, what was learned, and how learnings inform next steps.	Addresses design execution, analysis, and synthesis — the core iterative design cycle. Requiring teams to explain 'how findings inform next steps' establishes an explicit feedback loop between evidence and design decisions.
Execution (6 pts): The approach taken was rigorous, principled, and reflects professional HCD standards.	This criterion explicitly requires principled and rigorous execution — language that maps directly to ABET's requirement for systematic, evidence-based design. 'Professional HCD standards' establishes that design quality is assessed against an external disciplinary benchmark, not merely internal course expectations.
Presentation (3 pts): The report is clearly written and professionally presented.	Writing and communication quality.

Three milestones at 15 pts each constitute the full Project Process (P*) category at 25% of the course grade. These are 'execution' assignments, not planning — students are carrying out the research and design activities they specified in 492, generating evidence, and iterating based on what they find.

Primary Evidence: F4 — Process Book

The Process Book (50 pts) is the capstone artifact and the single highest-point assignment in the course. The Process Book Rubric assesses:

Criterion	Points / ABET Relevance
Concept Specification (20 pts): The process book clearly and fully articulates the project's design question, target users, and the problem being addressed.	Design requirements development: framing the problem, identifying users, and specifying the design challenge. Maps directly to ABET's 'developing requirements' and 'identifying opportunities' steps.
Design Specification (20 pts): The process book presents a clear design specification using appropriate techniques such as system diagrams, experience maps, user flow diagrams, or similar representations. Interaction design for the main components is specified.	This is the highest-evidence engineering design criterion in the curriculum. Requiring system diagrams, experience maps, user flow diagrams, and interaction specifications is explicitly engineering design output — the kind of design specification documentation that professional engineers produce. The rubric does not accept vague descriptions; it requires formal representations of the designed artifact.
Process Comprehensiveness (20 pts): The process book includes documentation of the complete design process with credible evidence. Credible evidence includes: user quotes, survey data, evaluation findings.	The explicit mention of survey data alongside user quotes and evaluation findings indicates that quantitative data is expected as part of credible evidence, even if not required from all teams. 'Complete design process' means students must document the full arc from initial framing through final specifications.
Process Narrative (10 pts): The process book tells a coherent, honest story of how the design developed over the course of the project.	Communicates the design rationale — documenting decision-making and how the design evolved in response to evidence.
Reflection (10 pts): The process book includes a thoughtful, critical reflection on the design process and outcomes.	Critical reflection on design decisions and trade-offs.
Presentation Quality (20 pts): The process book is professionally produced, well-organized, and visually clear.	Professional communication quality.

The Design Specification criterion (20 pts) is the most important single piece of evidence for the ABET self-study. Requiring system diagrams, experience maps, user flow diagrams, and interaction specifications for the main components establishes that this course produces formal engineering design artifacts, not merely narrative accounts of design activity. This is the same type of specification output that distinguishes engineering design from adjacent activities such as consulting, storytelling, or user advocacy.

Supporting Evidence: S1 and S2 — Status Reviews

The two status review assignments (S1, S2; 16 pts each) require teams to provide structured project updates. Each review includes: a summary of project progress, current milestone status, a link to an updated Gantt chart, identification of at least three project risks, and the team's current mitigation plan for each risk.

The risk identification and mitigation planning component maps to ABET's requirement that engineering design considers risks and makes trade-offs. It also reflects professional engineering project

management practice. Status reviews are weighted in the Admin category and are not pure engineering design, but they are integral to the iterative design process — not external to it.

Supporting Evidence: PF1, PF2, PF3 — Peer Feedback

Three rounds of structured peer feedback (PF1, PF2, PF3; 5 pts each) close the iterative critique loop. Teams review other teams' milestone work and provide feedback that informs subsequent design iterations. This pattern — iterative critique from informed peers followed by revision — is consistent with professional design review processes and with ABET's emphasis on iteration and evaluation.

P0 — Revised Project Plan

P0 (5 pts) asks students to update their 492 project plan at the start of the quarter, incorporating any changes to the project scope, team, or sponsor situation. It is a lightweight but real design planning activity — students are making explicit, reasoned decisions about their design approach before execution begins.

What Is Excluded

F2 (Project Video, 20 pts) and F3 (Showcase Event, 10 pts) are communication activities — documentation and public presentation of completed work. They demonstrate that design work was done but do not themselves constitute engineering design. F5 (Final Reflection, 5 pts) is individual reflective writing. These three assignments (35 pts, ~20% of the F* category total of 175 pts across all F assignments) are excluded from the engineering design credit estimate.

Credit Estimate Rationale

The low estimate of 3.0 QC is derived conservatively: P* (25% of grade = 1.25 QC) + F4 Process Book (~28.6% of grade = ~1.43 QC) + F1 Poster (~11.4% = ~0.57 QC) = 3.25 QC, rounded down to 3.0 to account for the fact that not all poster content constitutes engineering design.

The high estimate of 4.0 QC adds partial credit from the Admin category. S1/S2 status reviews (project management and risk analysis) and PF peer feedback (iterative critique) are integral to the design process, though the category also contains purely administrative items. The 4.0 QC high estimate treats approximately 2/3 of the Admin category as design-process activities. The full 5 QC course credit is not claimed because F2, F3, and F5 are excluded.

Confidence is High because the rubrics directly confirm the engineering design content claims. The Design Specification criterion in the Process Book Rubric is as explicit as any rubric criterion in the HCDE curriculum. No interpretive stretching is required to justify the design credit — the course delivers engineering design execution as its primary activity.

Engineering Science

Credit Estimate: 0.0–0.5 QC | Confidence: Low

Engineering science content in HCDE 493 is project-dependent and cannot be established from the course structure alone. The course does not prescribe specific research methods — teams execute whatever methods they specified in their 492 proposal. Some teams will use quantitative methods (survey

analysis, usability metrics, controlled evaluation); others may work entirely within qualitative traditions (interviews, observations, co-design sessions).

The Process Book Rubric's Process Comprehensiveness criterion mentions 'survey data, evaluation findings' alongside 'user quotes' as forms of credible evidence. This language suggests that quantitative evidence is expected and valued, even if not universally required. Teams using survey instruments or standardized usability scales (such as SUS, used in HCDE 303) would generate engineering science content through applied quantitative methods.

However, no rubric criterion in the export requires quantitative data collection, analysis, or reporting. The milestone rubric criteria (Completeness, Execution, Presentation) are method-agnostic. The Process Comprehensiveness criterion lists quantitative evidence as an example, not a requirement.

The 0.0–0.5 QC range spans this uncertainty. The low estimate (0.0 QC) reflects projects that use entirely qualitative methods. The high estimate (0.5 QC) reflects projects that include one or more phases with survey analysis or usability metric collection and analysis. Do not cite HCDE 493 for engineering science credit in the self-study without noting that science content varies by team and cannot be confirmed from the course export alone.

Modern Engineering Tools

The tools used in HCDE 493 are project-dependent. The course export does not prescribe specific tools; teams use whatever their design methodology requires. Based on the types of projects typical in the HCDE Capstone and the tools inventory from HCDE 492 and 303, the expected tool set includes:

- Figma or equivalent prototyping tools (interaction design specification, required by the Design Specification rubric criterion)
- User research tools: screen/session recording software, survey platforms (Qualtrics, SurveyMonkey, Google Forms), or equivalent
- Project management tools: Gantt chart software, Slack, Google Drive/Docs/Slides (inherited from 492)
- Presentation and publication tools: Adobe suite or equivalent for poster production

Figma (or a comparable interaction design tool) is the most defensible engineering tools claim: the Process Book Rubric's Design Specification criterion requires system diagrams, experience maps, and user flow diagrams — output types that in practice require professional design software. Survey platforms used for structured data collection also qualify.

Because tools are not prescribed, the tools inventory cannot be stated with certainty for all teams. In the self-study, tools credit should be framed around the Design Specification requirement (which implies prototyping/diagramming tools) and noted as varying by project for other tools.

Summary and Self-Study Guidance

Course	HCDE 493: Senior Capstone
Engineering Design (QC)	3.0–4.0 QC (High confidence)
Engineering Science (QC)	0.0–0.5 QC (Low confidence)
Total 5b Low (SC)	2.0 SC
Total 5b High (SC)	3.0 SC

Modern Tools	Figma or equivalent (required by Design Specification rubric); survey platforms and recording tools (project-dependent)
Rubrics available	Yes — Process Book Rubric and Milestone Deliverable Rubric both present in export

Key points for the ABET self-study:

- Present HCDE 492 and 493 as a unified two-quarter capstone sequence. The 492 evaluation covers design planning; 493 covers design execution. Together they constitute a complete, client-sponsored engineering design experience consistent with the EAC definition.
- The Process Book Rubric's Design Specification criterion (requiring system diagrams, experience maps, user flow diagrams, and interaction specifications) is among the most explicit engineering design evidence in the HCDE curriculum. It should be cited directly in the self-study.
- The Milestone Deliverable Rubric's Execution criterion — 'approach was rigorous, principled, and reflects professional HCD standards' — establishes a disciplinary quality benchmark for design execution. This language is valuable for ABET reviewers who question whether HCD methods constitute engineering practice.
- Engineering science credit is Low confidence and project-dependent. Do not cite 493 for engineering science without disclosing this variability. If specific capstone projects with quantitative methods are available as evidence, they can strengthen the case.
- The exclusion of F2 (video), F3 (showcase), and F5 (reflection) from the design credit estimate is intentional and conservative. These are communication and reflection activities, not engineering design. Reviewers will notice if these are counted, so the self-study should not count them.
- The 3.0 SC low estimate is the most defensible floor for this course. Even a strict reviewer who excludes poster work and Admin activities must credit P* milestone execution (1.25 QC) and F4 Process Book (1.43 QC) as qualifying engineering design content, yielding 2.68 QC (1.79 SC). The 2.0 SC floor reflects inclusive counting of F1 and modest Admin.

*_Last updated: April 13, 2026 — 4:30 PM_*